Epoch Ledger

Completed epochs through S3 e14: loss, ordinal agreement, minority-class recall, screening/STDR discrimination, hard-example-mining state, and per-grade recall, loaded from the saved training history.

EPOCHS COMPLETE	BEST QWK	BEST SCORE	BEST MIN-CLASS RECALL	BEST REFAUC
31 / 45	0.607	0.513	0.474	0.873
S4E1 950/3000 @33:45	stage 3, epoch 6	stage 3, epoch 13	stage 2, epoch 7	stage 3, epoch 6

Column glossary

Time — context

Wall-clock minutes to train the epoch; not a quality signal.

Loss — lower better

Weighted sum of every training objective, averaged per image.

QWK — higher better

Quadratic Weighted Kappa: ordinal agreement, penalises distant misses harder.

F1 — higher better

Macro F1 across all 5 grades, treating each equally.

Min-recall — higher better

Recall of whichever grade is doing worst that epoch - the key imbalance metric.

Score — higher better

Composite selection score used to pick the best checkpoint.

Accuracy — higher, cautiously

Overall correct-grade rate; can rise just by favouring grade 0.

refAUC — higher better

AUC for the binary referable-vs-not-referable DR split.

Screen cleared — context

Share of images passed without flagging; trades off against NPV.

Screen NPV — higher better

Negative predictive value of the screening gate vs a 0.985 target.

Screen thr — context

Decision threshold that hit the NPV target this epoch.

STDR AUC — higher better

AUC for detecting sight-threatening DR (Severe + PDR).

Miner p90 H — context

90th-percentile mining hardness carried into the next epoch.

Boundary rate — context

Share of samples flagged as one-grade boundary errors.

Per-grade recall — higher better, all 5

Recall per grade 0-4 - see the dedicated heatmap page.

Full comparison, all 31 completed epochs

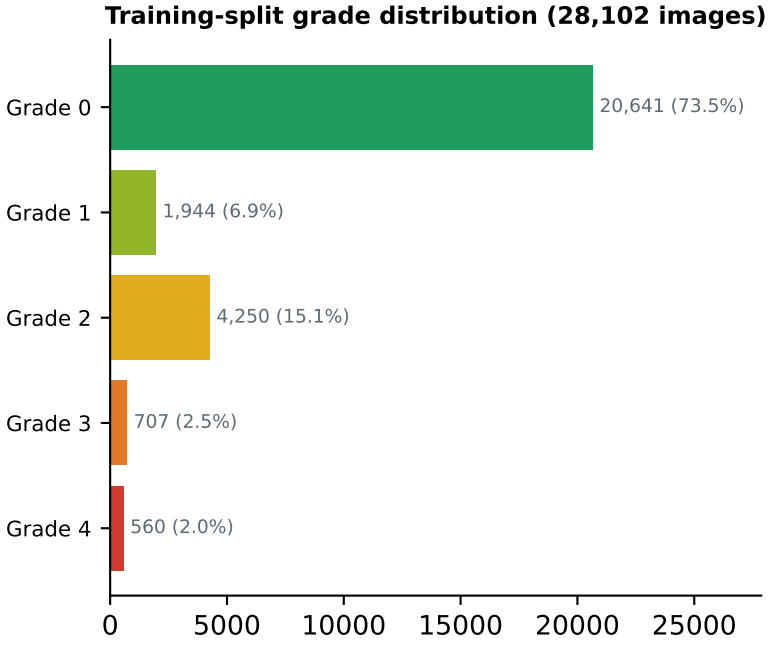
Stage 1: head only, backbone frozen. Stage 2: GLA-LoRA + ALPP trainable, backbone still frozen. Stage 3: 8/24 backbone blocks unfrozen too.
Shading: worst (red) to best (green) per column; loss inverted (lower is greener).

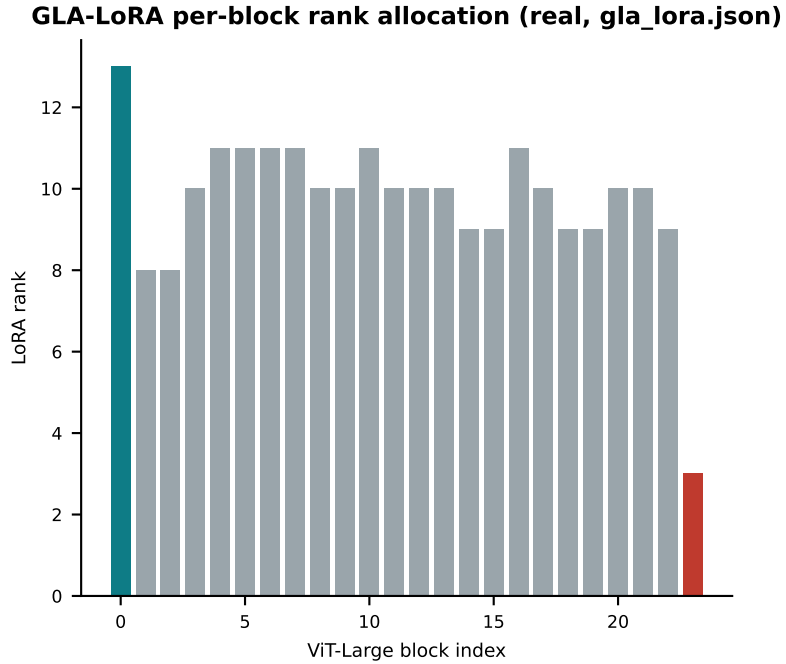
Epoch	Time	Loss	QWK	F1	Min-recall	Score	Accuracy	refAUC	Screen cleared	Screen NPV	Screen thr	STDR AUC	Miner p90 H	Boundary rate
S1 e1	75.7m	83.90	0.350	0.292	0.187	0.300	0.666	0.726	91.1%	0.9854	0.237	0.8877	1.00	0.03
S1 e2	68.8m	48.43	0.421	0.287	0.198	0.336	0.607	0.778	94.1%	0.9851	0.599	0.9162	1.00	0.05
S1 e3	75.6m	33.65	0.441	0.284	0.168	0.339	0.634	0.794	93.9%	0.9851	0.590	0.9165	1.15	0.06
S1 e4	75.6m	27.29	0.403	0.301	0.289	0.350	0.569	0.800	94.1%	0.9851	0.668	0.9149	1.28	0.07
S1 e5	70.1m	25.51	0.458	0.309	0.211	0.364	0.639	0.808	93.9%	0.9851	0.646	0.9176	1.37	0.08
S2 e1	83.4m	24.05	0.434	0.287	0.228	0.349	0.539	0.809	94.9%	0.9852	0.697	0.9203	1.43	0.08
S2 e2	95.0m	17.47	0.496	0.376	0.269	0.414	0.725	0.804	94.7%	0.9852	0.470	0.9163	1.47	0.08
S2 e3	85.7m	12.15	0.465	0.303	0.238	0.371	0.609	0.813	94.6%	0.9852	0.652	0.9201	1.52	0.08
S2 e4	98.8m	9.34	0.456	0.318	0.301	0.384	0.587	0.820	94.5%	0.9852	0.686	0.9215	1.55	0.09
S2 e5	100.8m	8.13	0.554	0.399	0.369	0.471	0.703	0.832	94.2%	0.9851	0.564	0.9461	1.58	0.09
S2 e6	91.5m	7.34	0.554	0.390	0.334	0.461	0.691	0.836	94.2%	0.9851	0.549	0.9345	1.59	0.09
S2 e7	98.4m	6.95	0.402	0.344	0.474	0.399	0.479	0.836	94.9%	0.9853	0.662	0.9459	1.61	0.09
S2 e8	91.0m	6.62	0.509	0.380	0.385	0.445	0.611	0.839	95.5%	0.9853	0.643	0.9413	1.61	0.09
S2 e9	96.6m	6.37	0.473	0.366	0.418	0.430	0.578	0.848	95.0%	0.9853	0.693	0.9426	1.60	0.09
S2 e10	83.5m	6.20	0.482	0.380	0.438	0.443	0.587	0.839	94.9%	0.9852	0.691	0.9410	1.60	0.09
S2 e11	92.2m	6.08	0.510	0.391	0.434	0.459	0.615	0.839	94.7%	0.9852	0.665	0.9499	1.61	0.09
S2 e12	101.9m	6.11	0.493	0.387	0.439	0.450	0.603	0.840	94.9%	0.9852	0.667	0.9461	1.61	0.09
S3 e1	212.7m	6.00	0.448	0.351	0.443	0.418	0.544	0.839	94.9%	0.9853	0.687	0.9549	1.61	0.09
S3 e2	194.1m	5.87	0.479	0.370	0.445	0.440	0.600	0.845	95.7%	0.9854	0.659	0.9532	1.61	0.09
S3 e3	214.3m	5.52	0.506	0.375	0.436	0.453	0.585	0.855	95.8%	0.9854	0.683	0.9613	1.61	0.09
S3 e4	217.1m	5.32	0.549	0.393	0.416	0.476	0.632	0.863	96.2%	0.9854	0.695	0.9645	1.60	0.10
S3 e5	222.1m	5.17	0.577	0.399	0.395	0.487	0.675	0.862	96.1%	0.9854	0.680	0.9657	1.60	0.10
S3 e6	302.4m	5.09	0.607	0.412	0.389	0.505	0.697	0.873	95.8%	0.9854	0.660	0.9628	1.61	0.10
S3 e7	14.0m	4.87	0.557	0.401	0.420	0.483	0.645	0.870	95.8%	0.9854	0.676	0.9640	1.61	0.10
S3 e8	48.3m	4.84	0.584	0.421	0.402	0.499	0.663	0.871	96.1%	0.9854	0.679	0.9622	1.61	0.10
S3 e9	32.9m	4.73	0.588	0.420	0.417	0.504	0.672	0.870	94.9%	0.9852	0.623	0.9593	1.61	0.10
S3 e10	116.6m	4.62	0.529	0.402	0.466	0.478	0.585	0.865	95.7%	0.9854	0.680	0.9576	1.60	0.10
S3 e11	226.2m	4.59	0.590	0.427	0.434	0.510	0.670	0.869	95.1%	0.9853	0.616	0.9595	1.60	0.10
S3 e12	48.4m	4.65	0.599	0.419	0.398	0.505	0.683	0.872	95.5%	0.9853	0.668	0.9604	1.60	0.10
S3 e13	40.4m	4.60	0.596	0.432	0.429	0.513	0.673	0.871	95.0%	0.9853	0.630	0.9561	1.61	0.10
S3 e14	56.4m	4.74	0.579	0.430	0.442	0.507	0.661	0.871	95.4%	0.9853	0.645	0.9592	1.60	0.10

Per-grade recall, every epoch

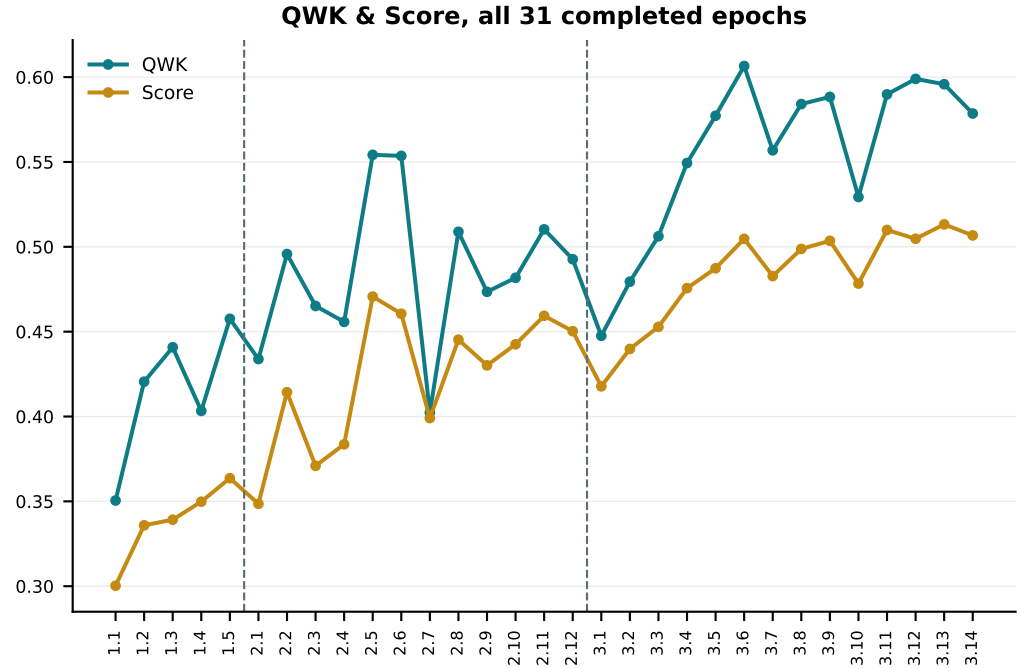
Shaded per grade (each column against its own best/worst across all 31 completed epochs) so each grade shows its own progress.

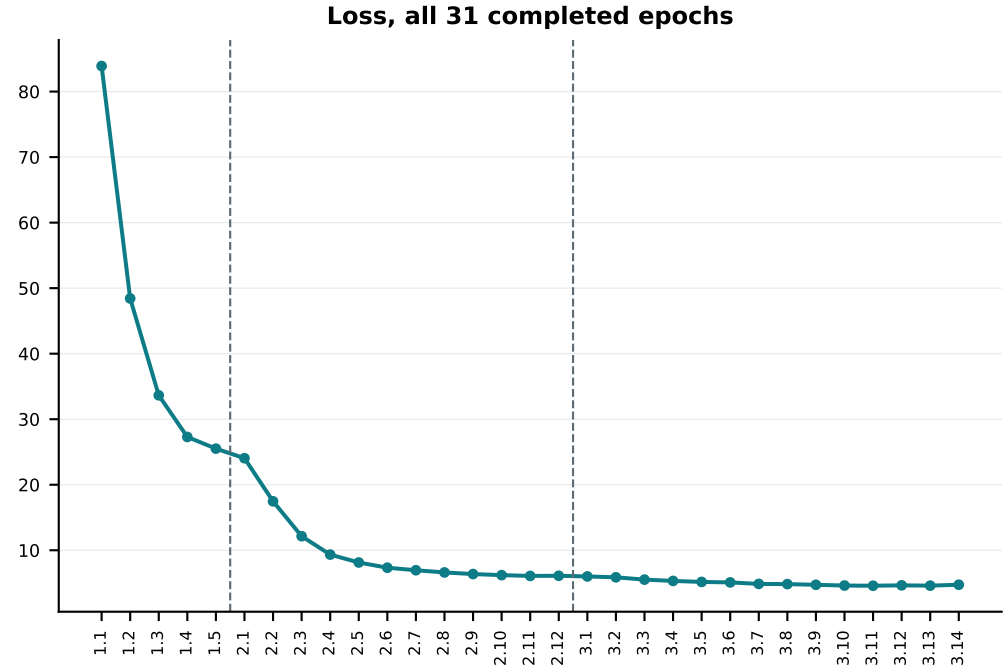
Epoch	Grade 0 No DR	Grade 1 Mild	Grade 2 Moderate	Grade 3 Severe	Grade 4 PDR
S1 e1	82.9%	5.9%	22.4%	27.8%	0.0%
S1 e2	72.6%	5.9%	34.1%	19.4%	28.6%
S1 e3	76.7%	6.9%	32.2%	11.1%	28.6%
S1 e4	65.3%	10.9%	39.7%	36.1%	33.3%
S1 e5	75.8%	5.9%	37.9%	19.4%	42.9%
S2 e1	61.1%	14.9%	42.5%	11.1%	42.9%
S2 e2	89.5%	5.0%	22.9%	52.8%	23.8%
S2 e3	71.9%	9.9%	36.4%	25.0%	28.6%
S2 e4	67.6%	10.9%	40.7%	38.9%	33.3%
S2 e5	83.5%	5.0%	36.4%	69.4%	23.8%
S2 e6	82.4%	5.0%	34.1%	61.1%	28.6%
S2 e7	48.9%	21.8%	53.7%	66.7%	28.6%
S2 e8	69.5%	9.9%	44.4%	61.1%	28.6%
S2 e9	63.9%	13.9%	47.7%	63.9%	33.3%
S2 e10	65.6%	12.9%	43.5%	75.0%	38.1%
S2 e11	69.5%	11.9%	43.5%	75.0%	33.3%
S2 e12	67.6%	12.9%	43.9%	75.0%	33.3%
S3 e1	60.2%	8.9%	43.5%	80.6%	28.6%
S3 e2	67.9%	11.9%	41.1%	80.6%	23.8%
S3 e3	65.9%	11.9%	41.1%	77.8%	28.6%
S3 e4	72.3%	10.9%	41.6%	72.2%	28.6%
S3 e5	79.1%	13.9%	35.0%	69.4%	28.6%
S3 e6	82.0%	10.9%	36.4%	69.4%	28.6%
S3 e7	73.8%	10.9%	43.0%	72.2%	33.3%
S3 e8	76.6%	10.9%	40.2%	69.4%	42.9%
S3 e9	77.9%	11.9%	38.3%	75.0%	38.1%
S3 e10	64.5%	21.8%	43.0%	75.0%	42.9%
S3 e11	77.0%	10.9%	41.6%	77.8%	38.1%
S3 e12	79.3%	9.9%	40.2%	69.4%	42.9%
S3 e13	77.0%	12.9%	43.5%	72.2%	42.9%
S3 e14	75.1%	16.8%	43.5%	72.2%	42.9%

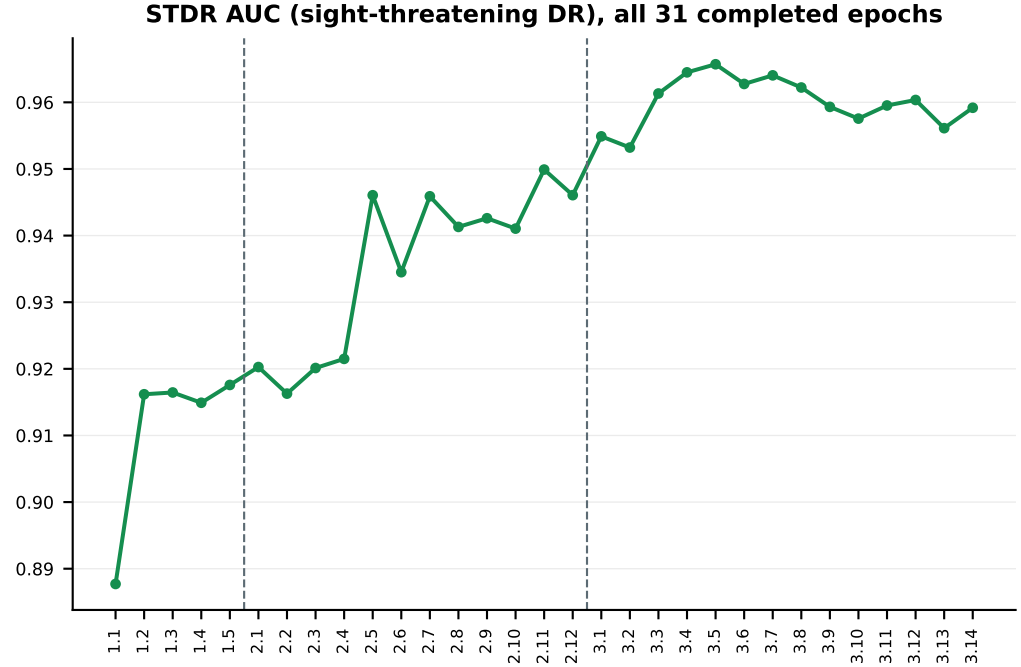


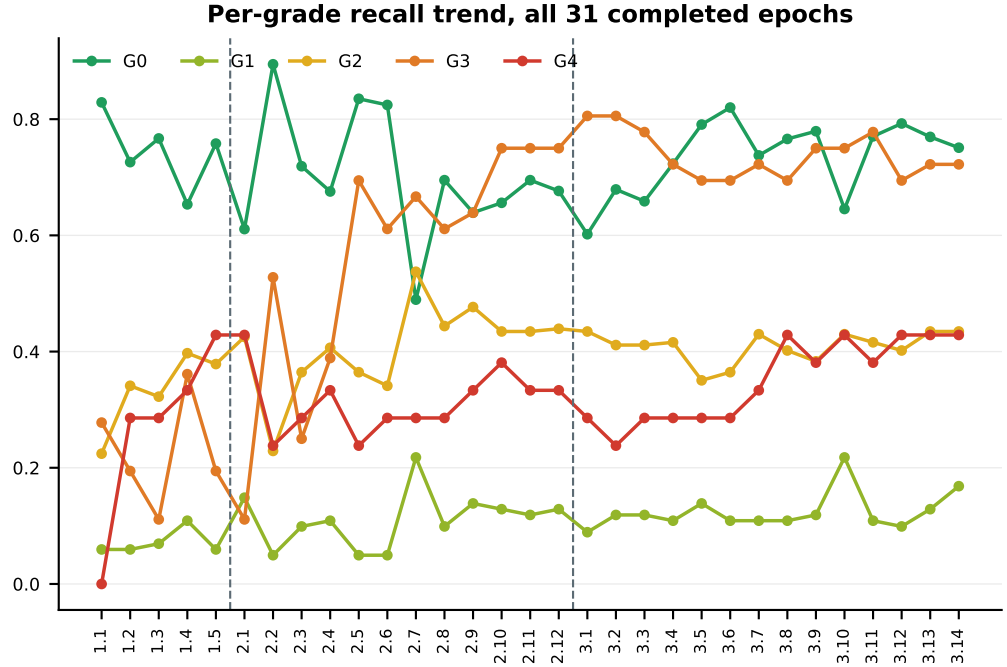


Trends


QWK & Score, all 31 completed epochs
QWK
Score
0.60
0.55
0.50
0.45
0.40
0.35
0.30
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11 3.12 3.13 3.14


Loss, all 31 completed epochs
80
70
60
50
40
30
20
10
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11 3.12 3.13 3.14


STDR AUC (sight-threatening DR), all 31 completed epochs
0.96
0.95
0.94
0.93
0.92
0.91
0.90
0.89
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11 3.12 3.13 3.14

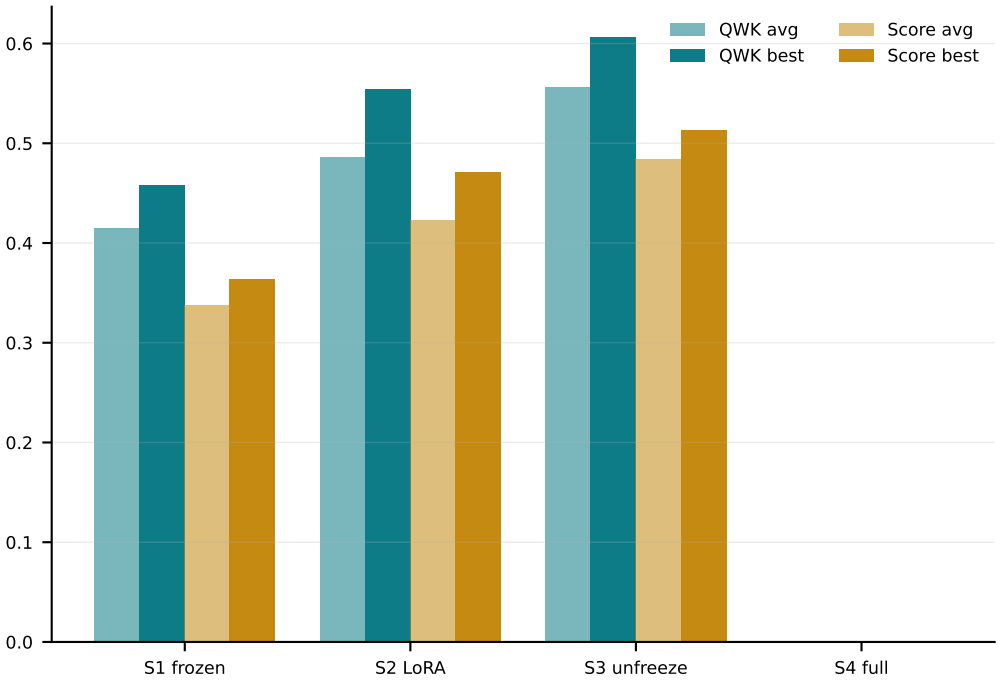

Per-grade recall trend, all 31 completed epochs
G0
G1
G2
G3
G4
0.8
0.6
0.4
0.2
0.0
1.1 1.2 1.3 1.4 1.5 2.1 2.2 2.3 2.4 2.5 2.6 2.7 2.8 2.9 2.10 2.11 2.12 3.1 3.2 3.3 3.4 3.5 3.6 3.7 3.8 3.9 3.10 3.11 3.12 3.13 3.14

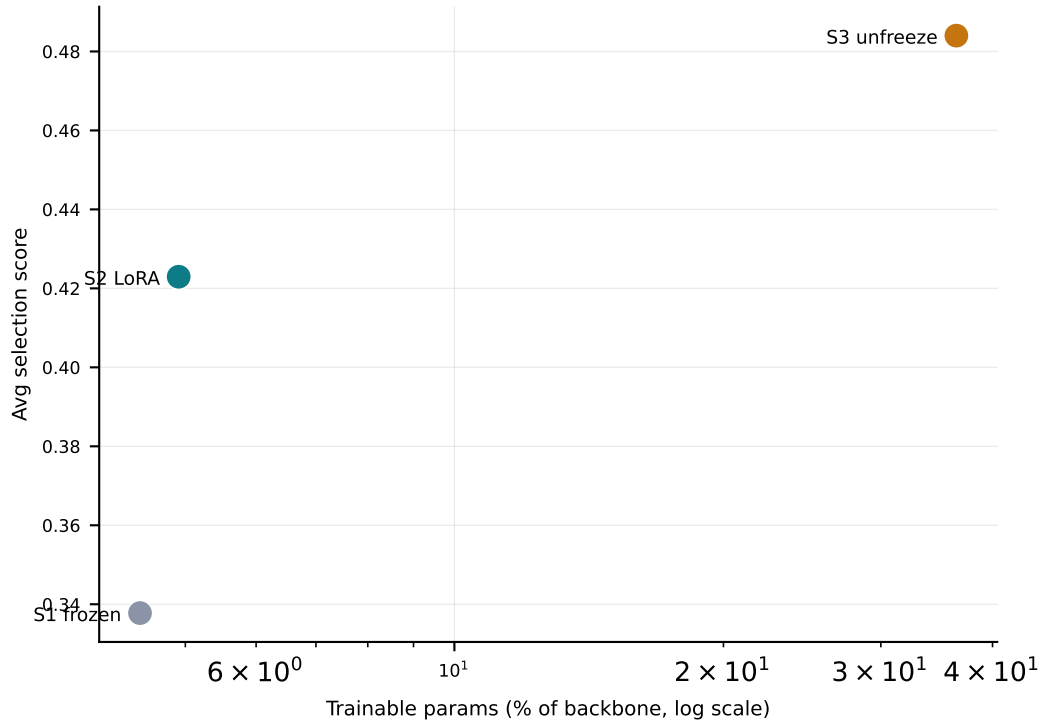
Stage-by-Stage Comparison

What each stage changed vs. what it achieved - S4 is config only, not yet run.

Stage	Epochs	Trainable params	% of backbone	Head / ladder LR	LoRA / backbone LR	Avg QWK	Best QWK	Avg score	Best score	Avg refAUC	Avg min-rec	Avg min /epoch
S1 frozen	5/5	14,220,579	4.45%	1.0e-4 / 1.0e-3	n/a / n/a	0.415	0.458	0.338	0.364	0.781	0.211	73
S2 LoRA	12/12	15,709,051	4.92%	1.0e-4 / 1.0e-3	5.0e-5 / n/a	0.486	0.554	0.423	0.471	0.830	0.361	93
S3 unfreeze	14/14	116,478,843	36.44%	5.0e-5 / 5.0e-4	5.0e-5 / 5.0e-6	0.556	0.607	0.484	0.513	0.864	0.424	139
S4 full	0/14	319,612,795	100.00%	2.0e-5 / 2.0e-4	2.0e-5 / 5.0e-6	-	-	-	-	-	-	-

Stage averages and per-epoch timings above are calculated from completed history rows; the latest monitor snapshot is shown separately from completed-epoch results.





Notable events

Most recent completed epoch

S3 e14: train loss 4.740, QWK 0.579, selection score 0.507, refAUC 0.871. These are validation metrics from the saved per-epoch history.

Live training snapshot

S4E1 950/3000 @33:45 The monitor records the most recent observed micro-batch step; it may lag behind a completed epoch while validation and checkpointing run.

Best recorded epoch so far

Best QWK: 0.607 (S3 e6); best selection score: 0.513 (S3 e13); best refAUC: 0.873 (S3 e6).

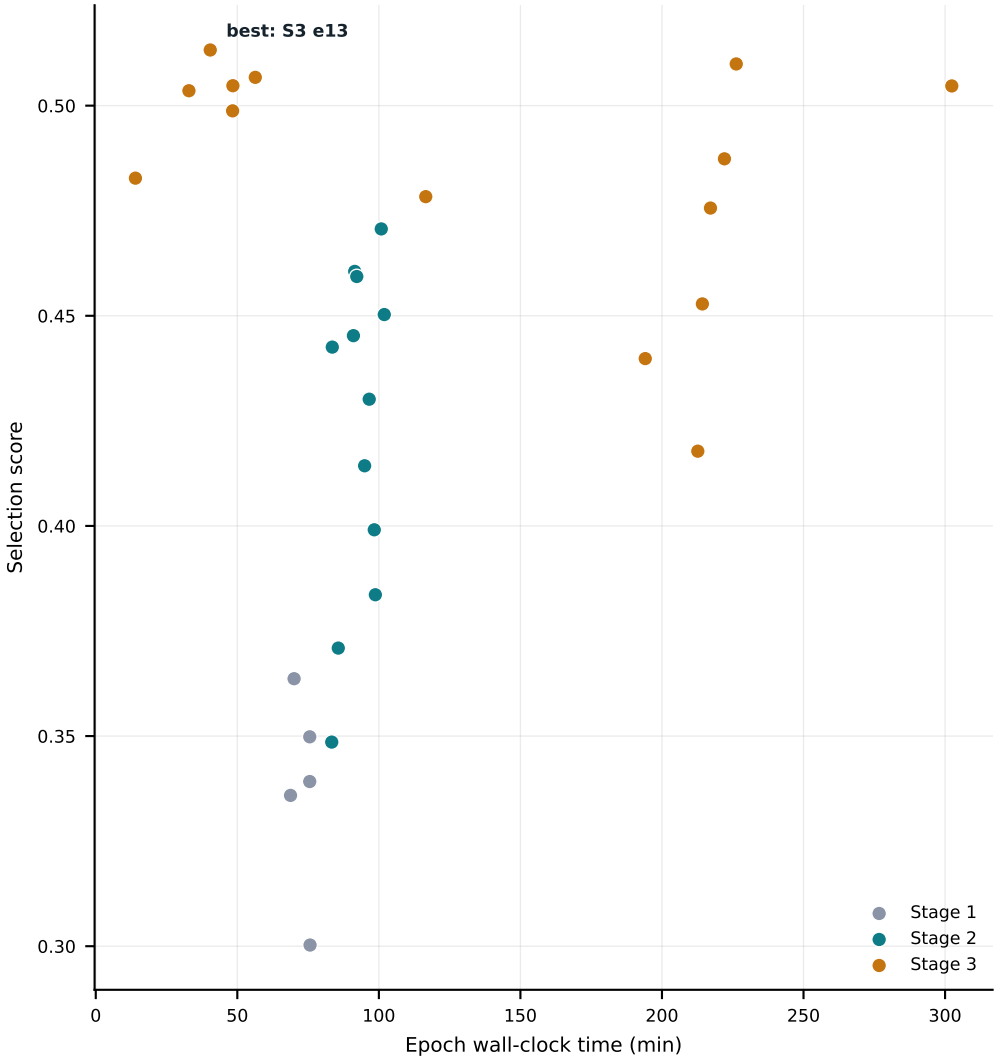
Setup & architecture reference

Dataset	35,126 images, 17,563 patients (EyePACS)
Split	28,102 / 3,512 / 3,512 train/val/test, no patient leakage
RETFound coverage	100% - 294/294 backbone tensors loaded
GLA-LoRA params	1,431,552 - 0.45% of the 319.6M backbone
Sampling ratio	5.01x grade4:grade0 weight, n^-0.5

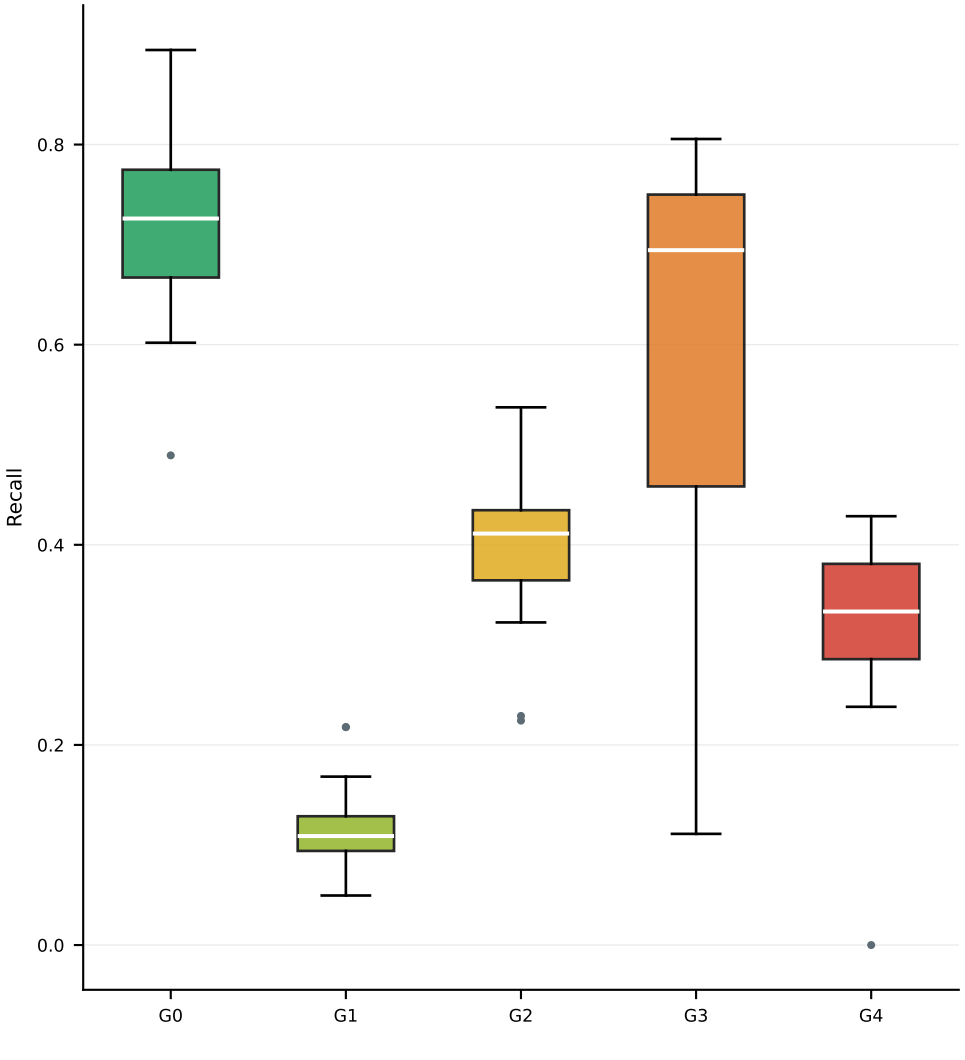
Efficiency & Spread

Two more views of the same 31 completed epochs: is the extra training time paying off, and how much does each grade's recall vary epoch to epoch.

Score vs. training time - is the extra time paying off?



Per-grade recall spread, all 31 completed epochs



Detailed Analysis & Conclusion

The current read on 31 completed epochs, alongside live run status (1 of 2).

1. Training progress

The saved history contains 31 completed epochs out of 45 planned. The latest completed result is S3 e14 with QWK 0.579, macro F1 0.430, selection score 0.507, and refAUC 0.871. The monitor's current snapshot is S4E1 950/3000 @33:45. An in-progress epoch is not counted among completed history rows.

2. Best observed validation metrics

The highest QWK is 0.607 at S3 e6; highest selection score is 0.513 at S3 e13; highest refAUC is 0.873 at S3 e6. These are maxima among the completed epochs in this history, not an independent test-set estimate.

3. Grade-level recall

The table and heatmap show validation recall for each grade in every completed epoch. Most recent grade recalls (grades 0-4): 75.1%, 16.8%, 43.5%, 72.2%, 42.9%. These validation values describe model behavior on this split; they should not be interpreted as clinical performance or deployment evidence.

Detailed Analysis & Conclusion

Current status and interpretation (2 of 2).

4. Screening metrics are validation results

The latest completed epoch reports screening NPV 0.9853, cleared 95.4% at threshold 0.645, and sight-threatening DR AUC 0.9592. These are validation metrics from the training run, not held-out test results or evidence of clinical safety.

5. Current run state

S4E1 950/3000 @33:45 This monitor snapshot is a separate, timestamped observation from the 31 completed epoch records. The next epoch's final metrics will appear only after training, validation, and checkpointing finish.

6. Reading the ledger

Epoch-level plots, tables, stage comparisons, and grade-recall summaries are generated from the saved history CSV. Fixed architecture and split details are reference information. The ledger reports observed training-run results only; assess generalization using the separate held-out evaluation outputs.